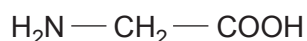


11. (a) Aminoethanoic acid is the simplest α -amino acid.



- (i) Give the structure of the dipeptide formed from two molecules of aminoethanoic acid. [1]

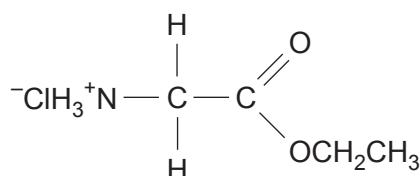
- (ii) Explain why aminoethanoic acid can only form one dipeptide. [1]

.....

.....

- (b) Amino acids form esters in the usual way from the carboxylic acid group and an alcohol, in the presence of an acid catalyst. These esters can form a salt with the acid used in esterification.

The formula of one of these salts is shown below. Compound **M** is a white solid that is soluble in water, giving a colourless solution.



Compound **M**

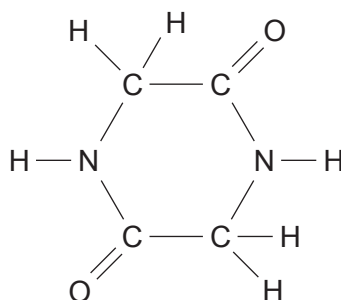
- (i) I. Many amino acids exist as zwitterion forms in aqueous solution.
Give the zwitterion form of aminoethanoic acid. [1]

- II. Explain why compound **M** cannot form a zwitterion in this way. [1]

.....



- (ii) Under suitable conditions, aminoethanoic acid condenses to give diketopiperazine.



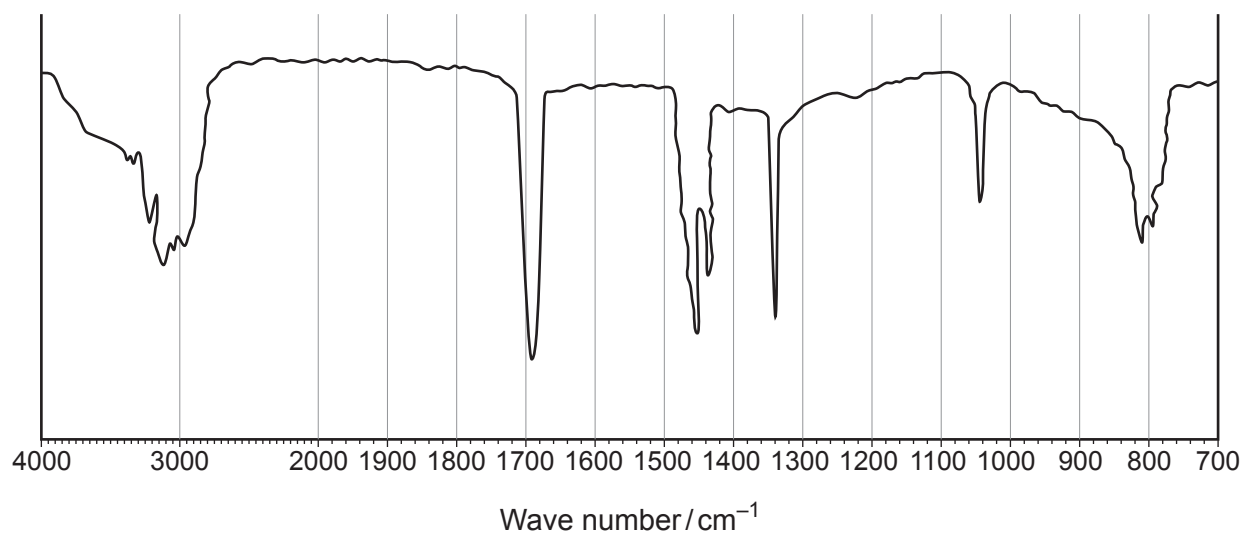
- I. The mass spectrum of diketopiperazine shows a molecular ion at m/z 114 and two prominent fragments at m/z 43 and 71.

Suggest a formula for the fragment at m/z 71.

[1]

- II. An outline infrared absorption spectrum of diketopiperazine is shown below. Use the formula shown and the data sheet to suggest an absorption value for the N — H and C = O bonds.

[1]



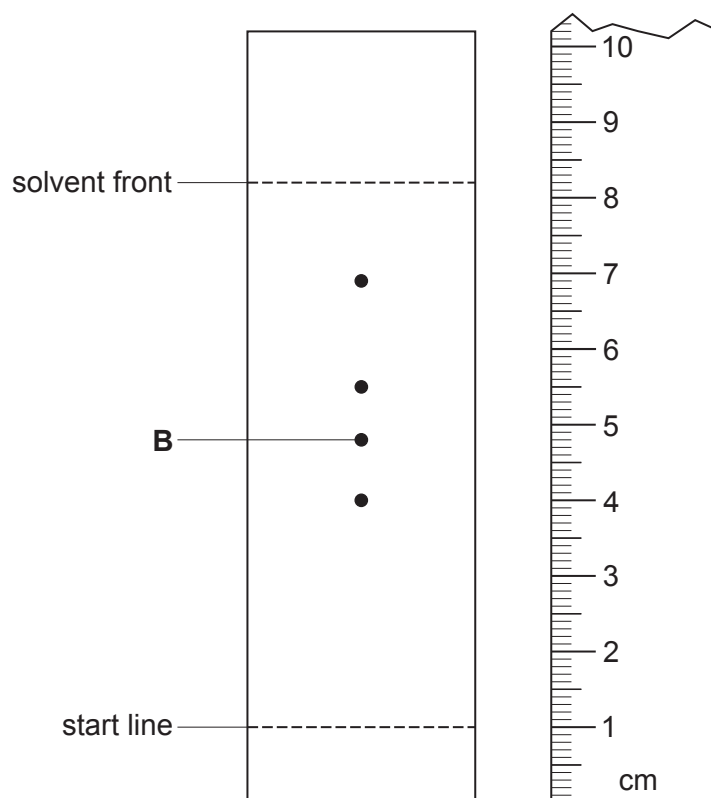
N — H cm^{-1} C = O cm^{-1}



(c) A mixture of α -amino acids can be separated and identified by thin layer chromatography.

- (i) These amino acids are colourless and the chromatogram is sprayed with a solution of ninhydrin, so that the amino acids appear as purple dots. The colour is due to the dye Ruhemann's Purple.

A thin layer chromatogram of a mixture of α -amino acids is shown below.



Calculate the R_f value for threonine (spot **B**).

[1]

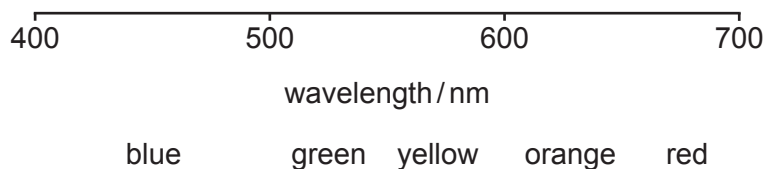


- (ii) Ruhemann's Purple has a maximum absorption at 564 nm.

The information below indicates the colours absorbed at various wavelengths in the visible spectrum.

Use this to help you explain why the colour of the dye is purple.

[1]



.....

.....

.....

- (iii) Calculate the energy (in kJ mol^{-1}) associated with this absorption at 564 nm.

[3]

energy = kJ mol^{-1}



(d) Amino acids react with nitric(III) acid to produce nitrogen gas.



- (i) 0.500 g of 2-aminohexanoic acid, $\text{CH}_3(\text{CH}_2)_3\text{CH}(\text{NH}_2)\text{COOH}$, reacted with an excess of nitric(III) acid.

Calculate the volume of nitrogen produced, assume the temperature was measured at 298 K and at 1 atmosphere pressure.

[2]

volume of nitrogen produced = cm^3

- (ii) The actual volume of nitrogen produced was 90.9 cm^3 , which was less than the calculated volume in part (i) above.

Suggest **two** reasons for this low result, apart from errors in weighing and in measuring the volume of nitrogen produced.

[2]

1.

.....

2.

.....

